

Brightline Research Assignment 2

01 Foundational Concepts of Analytics

01) Business Analytics is the use of data, statistics and analytical tools to understand how a business is performing and to help solve business problems. It helps businesses use information to make better decisions and find opportunities for improvements. Business analytics is broader than simply analysing data because it focuses on applying the analysis to a specific business problem or business decision.

Example: a clothing store can analyse its sales data to see which products sell the most and use this information to decide which product to order more of.

Source: IBM. (2023). What is Business Analytics? IBM.

02) Data Analytics is the process of examining data to find useful information, patterns and answer to specific questions. It can be used in many different fields, not just only business, to answer questions and support decisions. Data analytics focuses mainly on examining data. It is not necessarily focused on a business problem, whereas business analytics specifically applies analysis to business decisions and problems.

Example: a supermarket could analyse its sales data to find out which product sell more of on weekends than during the week.

Source: IBM. (2023). Data Science vs Data Analytics: What's the difference? IBM.

03) Business Intelligence is the use of processes and tools to turn business data into useful information that can be easily understood. It commonly uses reports, dashboards and visualisations to help businesses monitor performances and make informed decisions. I would say that the key distinction between data analytics, business analytics and business intelligence, is that data analytics focuses on what the data tell us, business analytics focuses on what can we do with the analyses to improve the business and business intelligence focus on what is happening in the business with the use of reports and dashboards.

- 3) Example: A company can use Power BI to create a dashboard showing its monthly sales, expenses and profits so that managers can quickly see how the business is performing.

Source: Microsoft. (n.d.). What is Business Intelligence? Microsoft Power BI.

- 4) Data-driven decision making means using facts and information from data to help make business decisions instead of relying only on guesses or opinions. Business can use data to understand past performances, identify problems and make better-informed choices.

Example: If a restaurant realizes from its sales that a certain meal sells poorly, it might remove the meal from its menu and replace it with customers would be more inclined to buy.

Sources: Microsoft (n.d.). What is Power BI? Microsoft Learn.

- 5) Big data refers to very large and complex amounts of data that can come from many different sources. Because of its size and complexity, businesses often need specialized tools and systems to store, process and analyse it.

Example: An online retailer like Takealot can collect huge amounts of data from customer searches, purchases, website activity and product reviews and analyse it to understand customer behaviour.

Source: IBM. (2025). What is Data? IBM

- 6) Descriptive Analytics look at historical data to explain what has already happened. It summarizes information such as sales, website visits or customer numbers so that business can understand its past performance.

b) Example: a business could look at its monthly sales reports and see that sales has increased by 15% during December.

Source: IBM.(2025). What is Business Analytics? IBM

7) Diagnostic Analytics looks at data to understand why something happened. It goes beyond simply reporting the results by looking for patterns, relationships and possible causes behind it.

Example: If a company notices that sales dropped in March, it could analyse sales by product, location and customer group to find out what caused this decline.

Source: IBM.(2025). What is Diagnostic Analytics? IBM

8) Predictive analytics uses historical and current data to identify patterns and estimate what could happen in the future. Businesses can use it to forecast things such as sales, customer behaviour and demand.

Example: a supermarket could use previous sales data to predict demand for cold drinks will increase during a hot weekend and order more stock.

Source: IBM(2025) What is Predictive Analytics. IBM

9) Prescriptive analytics goes one step further than predicting what might happen by using data to suggest what a business could or should do next. It can compare possible actions and help businesses choose action that could improve a particular outcome.

Example: a delivery company could use prescriptive analytics to recommend the most efficient delivery routes based on traffic, delivery times and available vehicles.

Source: IBM.(2025) What is Prescriptive Analytics? IBM

10) Data literacy is the ability to understand, work with and communicate information from data. A data-literate person should be able to understand what data means and use it appropriately when making decisions.

Example: a manager who can read a sales dashboard, understand the trends and use the information to make decisions about stock levels is demonstrating data literacy

Source:

02: Data, Measurement & Reporting

11) Structured data is information that is organised in a fixed format like rows & columns. It is easy to store, search and analyse because each piece of data has a specific place.

Example: a customer database with columns for customer name, age, email address, and purchase amount.

Source: Amazon Web Services (AWS). (2025) What is structured data? AWS.

12) Unstructured data is information that does not follow a fixed structure like rows & columns. It can include things such as texts, images, audio and videos, which usually need more advanced methods to analyse.

Example: customer reviews, photographs, whatsapp messages, videos or recorded phone calls.

Source: Amazon Web Services (AWS). (2025) What's the difference between structured data and unstructured data?

13) A data warehouse is a central place where a business collects and stores organised and cleaned data from source like a database so that it can be analysed and used for reporting and decision-making.

Example: a retail company could combine its sales, customer and inventory data in one data warehouse to analyse its overall performance

Source: IBM (2025). What is Data Warehouse? IBM

14) Data Mining is basically digging through data to find useful information and patterns that can help a business make better decisions.

Example: a supermarket can analyse thousands of customer transactions and discover that customers who buy nappies are also more likely to buy baby wipes. The supermarket can use this pattern to place the nappies and wipes closer together or create a promotion for both.

Source: IBM (2025) What is data mining? IBM

15) Data visualization is the process of turning data into visual formats such as charts, graphs and other visuals so that information is easier to understand and patterns can be spotted.

Example: a business could use a line graph to show how its sales have changed over the past year

Source: Tableau (nd) What is Data Visualization? Definition, Example, and Learning Resources.

16) A dashboard is a visual display that brings important information together in one place, allowing a business to monitor performance and see changes in its data.

Example: a sales dashboard could show today's sales, monthly revenue, no. of customers.

Source: Tableau (nd) What is a dashboard? A complete overview.

17) A metric is any measurable value used to track or understand something about a business. It can measure things like sales, website visits, nr of customers or delivery time.

Example: a company records 5800 website visitors in a month. The number of website visitors is a metric because it is a measurable value.

Source: IBM.(nd). Key technical concepts. IBM

18) A KPI is a measurable value that is specifically used to show whether a business is achieving an important goal. Therefore, every KPI is a metric, but not every metric is a KPI.

Example: If a company's goal is to increase website traffic, it could set 10000 website visitors per month as a KPI. The number of website visitors is now important because it is being used to measure progress towards a specific business goal.

Source: IBM.(nd.) Key Performance Indicators. IBM

19) A benchmark is a reference point used to compare performance. It helps a business whether its current results are better, worse or similar to a previous target or industry standard ^{see}

Example: If the delivery time for a specific industry is 2 days, a company can use 2 days as a benchmark to compare its own delivery performance.

Source: Microsoft learn (n.d) FinOps Benchmarking. Microsoft learn

20) Segmentation is the process of dividing a larger group of customer or data into smaller groups. The division into smaller groups are based on shared characteristics or attributes.

Example: a clothing company could segment its customers by age, location, spending habits or the types of products they buy.

Source: IBM.(nd). The customer segmentation solution plan. IBM

03: Customer & Sales KPIs

21) Conversion rate is the percentage of people who take a desired action after interacting with a business. The action could be buying a product, signing up for a service, filling in a form, or downloading something. It basically shows how many people actually did what the business wanted them to do.

Example: If 1 000 people visit an online store and 60 of them buy something, the conversion rate is 6%. This means that out of every 10 visitors, 6 become customers.

Source: Hubspot. (2020). Customer Acquisition Analytics: What to measure, where to optimize, connect it to revenue. Hubspot.

22) Customer Acquisition Cost (CAC) is the average amount a business spends to gain one new customer. It can include costs such as advertisements, sales salaries and marketing expenses.

Example: If a company spends R20 000 on sales and marketing and gains 100 new customers, its CAC is 200 per customer.

$$CAC = \text{total sales and marketing costs} \div \text{number of new customers.}$$

Source: Hubspot. (n.d) How to reduce Customer Acquisition Costs: 21 Proven Strategies. Hubspot.

23) A Customer Lifetime Value (CLV) is the estimated amount of revenue or profit a business can expect from a customer over the entire time they remain a customer. Instead of only looking at one purchase, CLV looks at the customer's relationship with the business over time.

Example: If a customer spends R500 every month and stays with the business for 2 years, the business can estimate the customer's lifetime value by looking at how much they are expected to spend during those two years.

The CLV:CAC ratio compares the value a business expects to get from a customer over the lifetime with the amount it cost to acquire that customer. It helps a business to see whether the money it spends to attract customer is likely to be worthwhile.

Example: If a customer has a CLV of R3000 and it costs the business R1000 to acquire that customer. Its CLV:CAC ratio is 3:1

This means that for every R1 the company spends to get this customer, the customer is expected to generate R3 in their lifetime value.

A healthy relationship would be a ratio above 1:1.

Source: Hubspot (2021) How to reduce Customer Acquisition cost: 21 Proven Strategies. Hubspot.

24) Churn rate is the percentage of customers who stop using a company's product or services during a specific period.

Example: If a subscription business starts the month with 1000 customers and 50 cancel its churn rate is $\frac{50}{1000} = 5\%$

Churn rate = $\frac{\text{lost customers}}{\text{customers at the beginning}} \times 100$

Source: Hubspot (2020). Customer Acquisition Analytics: What to measure, Where to optimize, connect it to revenue. Hubspot

25) Customer retention rate is the percentage of customers that a business keeps over a specific period. It is closely related to the churn because churn measures customers who leave, while retention measures customer who stay.

Example: If a company keeps 950 out of 1000 existing customers over a period, its retention rate is 95%

Source: Hubspot. (2025). Customer Retention Metrics: How to measure and improve retention. Hubspot

26) Net Promoter Score (NPS) is a measure of how likely customers are to recommend a company's product or services to other people. Customers are normally asked to rate how likely they are to recommend the company on a scale of 0-10. Their responses are then used to calculate the NPS.

Example: a company could ask its customer: "On a scale from 0 to 10, how likely are you to recommend us to a friend or colleague?"

Source: Hubspot (2025). Customer Retention Metrics: How to measure and improve retention. Hubspot

27) Average Order Value (AOV) is the average amount of money customers spend each time they place an order.

Example: if a store makes R10 000 from 200 orders its AOV is $\frac{10\ 000}{200} = 50$
So the average customer spends R50 per order.

$$AOV = \text{total revenue} \div \text{number of orders}$$

Source: Shopify (2026) Ecommerce Marketing guide for Online stores. Shopify

28) Sales funnel is the different stages a potential customer goes through before becoming a paying customer. It usually starts with awareness and moves through interest and consideration before ending with a purchase.

Example: a customer might: see a Facebook advert → visit the website → look at the product → add it to their cart → buy the product.

Not everyone who enters the funnel reaches the final stages, hence conversion rates.

Source: Hubspot (2026) Customer acquisition analytics: What to measure, where to optimize and how to correct revenue. Hubspot

29) A marketing campaign is a planned set of marketing activities that work together to achieve a specific goal, such as increasing sales, launching a product, entering a new market or generating awareness.

Example: A restaurant wants to attract more customers during its quiet weekday evening. It launches a campaign offering 20% off meals from Monday to Wednesday. The restaurant promotes the offer through Instagram posts, WhatsApp messages and email, and tracks how many people use the discount. This is a marketing campaign because all these activities are planned together to achieve one goal: increase weekday sales.

The campaign is what the business runs, while the other terms help measure whether the campaign worked

Source: Hubspot (n.d) How to create a successful marketing campaign Hubspot

30) Post-campaign analysis is the process of looking at the results of a marketing campaign after it has finished. The business checks things like sales, clicks, leads, revenue and costs to see what worked and what could be improved.

Example: After an Instagram campaign ends, a business checks how many people clicks the ads, how many bought a product or services and how much revenue was generated to determine whether the campaign was successful.

Post campaign analysis brings together measures such as CTR, conversion rate, incremental revenue, ROI and ROAS to evaluate the campaign.

Source: Hubspot (n.d) Analyze performance across campaigns in marketing studio. Hubspot

31) Incremental revenue is the extra revenue that was generated because of a specific campaign or marketing activity, above what the business would normally have earned. Total revenue is everything the business earned. Incremental revenue is the extra amount linked to the campaign.

Example: A business normally makes R50 000, in a month, but after running a campaign it makes R65 000. The 15 000 difference can be considered incremental revenue if the campaign caused that extra sales.

Source: Shopify (2025). Incremental sales: definition, formula, and tips. Shopify

32) ROI (Return on Investment) measures how much profit a business makes compared with what it spent on an investment. In marketing, it helps show whether a campaign made enough money to justify its overall costs.

$$ROI = (\text{Return} - \text{Investment costs}) \div \text{Investment Cost} \times 100$$

ROI looks at the overall profitability of the investment, not just advertising revenue.

Example: If a company spends R10 000 on a campaign and makes R30 000 in revenue, the basic return is R20 000 after the campaign cost, giving an ROI of 200%.

Source: Shopify (2025). Marketing ROI: How to calculate it (2025). Shopify

33) Return on Ads Spend (ROAS) measures how much revenue a business generates for every rand or dollar spent specifically on advertising. $ROAS = \text{Revenue from ads} \div \text{Advertising spent}$. ROAS focuses specifically on advertising revenue and ad spend, while ROI looks at the overall profitability of the investment. A campaign can have a high ROAS but still have a low ROI if other costs are high.

Example: If a company spends R5 000 on Google Ads and generates R20 000 in revenue, its ROAS is 4:1. This means it generates R4 in revenue for every R1 spent on advertising.

Source: Shopify (2025) ROAS vs ROI: how to calculate and use each metric. Shopify

34) Click-through rate is the percentage of people who see an advertisement, email or link and actually click on it. $CTR = \text{Clicks} \div \text{Impression} \times 100$
CTR helps show whether an advert or message is getting people's attention, but it does not tell you whether those people eventually bought something.

Example: If an advert is shown 10 000 times and receives 300 clicks, the CTR is 3%.

Source: Hubspot (nd) The essentials marketing & sales metrics your team should track. Hubspot

35) Bounce rate measures the percentage of website sessions where visitors leave without becoming meaningfully engaged with the website.
A campaign might have a good CTR because people are clicking the advert, but if many of those visitors leave the website without engaging, the campaign needs improvements.

Example: If 1000 people visit a website and 400 of their sessions are not engaged, the bounce rate would be 40%.

Source: Google Help (nd) Engagement rate and bounce rate. Google Help.

36) A/B testing is a method of comparing two versions of something to see which one performs better. Usually, one element is changed while the rest stays the same.
A/B testing can help improve metrics such as CTR, conversion rate and sales. The results can then be used in future campaigns.

Example: An online store creates 2 versions of an advert. Version A says "Shop Now" and version B says "Get 20% off." The business shows both versions to different groups and compares their results.

Source: Hubspot (nd) A/B testing your website - all the Hqs Devs and marketers need to know.
Hubspot

37) Lead generation is the process of attracting people who may be interested in a product or service and getting their information so the business can follow up with them. Lead generation helps fill the sales funnel with potential customers. These leads can then be nurtured until some of them become paying customers.

Example: a company offers a free guide on its website. Visitors enter their name and email address to download it. These people become leads that the company can contact later.

Source: Hubspot (nd) lead generation strategies to attract business and drive growth. Hubspot

38) Marketing attribution is the process of identifying which marketing channels or interactions contributed to a customer taking an action, such as making a purchase. Attribution helps businesses decide whether to give credit for results and can be used when calculating campaign revenue and ROI.

Example: A customer first sees a company on Instagram, later clicks a Google search advert and finally buys after receiving an email. Attribution helps the business understand how these different channels contributed to the purchase.

Source: Hubspot (nd). What is marketing attribution & how do you report on it? Hubspot.